

1. Business Context

In the financial services sector, consumer complaints serve as a direct indicator of product friction, service bottlenecks, and regulatory compliance risks. Organizations process tens of thousands of consumer disputes across various product lines—such as checking accounts, mortgages, credit reporting, and student loans.

Without structured, data-driven analysis, the raw volume of these complaints obscures systemic issues, limits the ability of the support operations team to allocate resources effectively, and increases the risk of regulatory penalties due to delayed resolution timelines.

2. The Core Problem

The organization currently lacks actionable, centralized visibility into its customer complaint pipeline. There is no clear understanding of which financial products drive the highest volume of disputes, the root causes of these issues, or whether the company's resolution process meets timely compliance standards across all regions.

3. Key Objectives

The primary objective of this analytics project is to ingest, clean, and analyze historical consumer complaints data to build a dynamic reporting environment. The analysis aims to answer the following critical business questions:

- **Product Friction:** What are the most complained-about products, and what specific issues (e.g., account management, incorrect reporting) are driving this volume?
- **Operational Efficiency:** How efficient is the organization at resolving complaints? This involves measuring the processing lag (time between submission and receipt) and tracking the percentage of timely responses.
- **Geographical Distribution:** Are there specific state-level hotspots experiencing disproportionately high complaint volumes that require targeted regional support?
- **Resolution Outcomes:** What types of resolutions (e.g., closed with monetary relief vs. closed with explanation) are most common, and how do these outcomes correlate with the product involved?

4. Analytical Approach

To solve this business challenge, the project will be executed through a structured data analytics workflow, culminating in an interactive business intelligence dashboard.

Phase 1: Data Preparation & Feature Engineering

Assess data quality by handling missing values (Nulls) in critical dimensions such as 'Sub-issue' and 'Sub-product'. Standardize date formatting and create custom calculated metrics, specifically utilizing DAX to measure the **Average Processing Lag** (the day differential between 'Date Submitted' and 'Date Received').

Phase 2: Exploratory Data Analysis (EDA)

Aggregate the dataset to identify macro trends. This includes isolating seasonal peaks in complaint volume, evaluating the distribution of complaints across various submission channels (Web, Phone, Postal Mail), and mapping out state-level operational efficiency.

Phase 3: Dashboard Development & Visualization

Develop an interactive, multi-page operational dashboard utilizing Microsoft Power BI. The dashboard will feature a premium, custom-designed dark theme to ensure maximum readability and professional presentation. It will include:

- An **Executive Overview** for tracking top-level KPIs (Total Complaints, Timely Response %).
- A **Geographical Analysis** view utilizing interactive filled maps to identify regional hotspots.
- A **Root Cause Analysis** page detailing issue breakdowns and resolution outcomes by product.
- A **Drill-Down Matrix** utilizing AI Key Influencers to diagnose the specific variables causing delayed responses.

5. Project Scope & Deliverables

Interactive Power BI Dashboard (.pbix)	Executive Insights Report	Data Cleaning Workflow
DAX Measure Library		